

Assignment 23 C++

Section - A

Q.1 Explain the constructor method. How it differs from other member function.

Ans A constructor is a special member function of a class that is automatically called when an object is created. It is used to initialize data members.

Differences from normal functions :

- constructor name = class name
- No return type (not even void)
- Called automatically
- used for initialization, not general tasks.

Q.2 Differentiate between default and parameterized constructor?

Ans	Feature	Default Constructor	Parameterized Constructor
	Arguments	No arguments	Takes arguments
	Purpose	Initialize with default values	Initializes with user-defined values
	Example	Student()	Student(int r, string n)

Q.3 Why is constructor used under public access specifier?

Ans Constructor are declared public so that objects of the class can be created from outside the class.
If private, object creation would be restricted.

Q.4 Complete below code :

```
Ans #include <iostream>
using namespace std;

class Student {
    int roll_number;
    string name;

public:
    // initializer list
    Student () : roll_number (1), name ("Aditya") {}

    void display () {
        cout << "Roll: " << roll_number << endl;
        cout << "Name: " << name << endl;
    }
};

int main () {
    Student s1;
    s1.display();
    return 0;
}
```

Q.5 what do you understand by initialization of object?

Ans Object initialization means assigning initial values to data members when an object is created, either using constructors or and initializer lists.

Section - B.

Q.1 write a program to depict the function of unary operator overloading.

Ans

```
#include <iostream>
using namespace std;

class Number {
    int x;
public:
    Number (int a) { x = a; }

    void operator - () {
        x = -x;
    }

    void display () {
        cout << x << endl;
    }
};

int main () {
    Number n(5);
    -n;
    n.display ();
    return 0;
}
```


Q.2 Write a program to implement binary operator overloading?

Ans

```
#include <iostream>
using namespace std;

class Number {
    int x;
public:
    Number (int a) { x = a; }

    Number operator + (Number n) {
        return Number (x + n.x);
    }

    void display () {
        cout << x << endl;
    }
};

int main () {
    Number n1(5), n2(3);
    Number n3 = n1 + n2;
    n3.display ();
    return 0;
}
```

Q.3 Differentiate between a Constructor and a Destructor.

Ans A constructor is a special member function used to initialize objects, whereas a Destructor is used to destroy objects and free resources.

Basis	Constructor	Destructor
Purpose	Initializes object	cleans up object
Name	Same as class name	Same as class name with ~
Return Type	No return type	No return type
Parameters	Can take parameters	cannot take parameters
Overloading	Can be overloaded	cannot be overloaded
Execution Time	Called when object is created	Called when object is destroyed
Memory Handling	Allocates / initializes	Deallocates memory

Section - C

Q.1 What is function overloading? What are the advantages of function overloading? Write a program to calculate the area of circle, area square and area of triangle using function overloading.

Ans Function overloading means defining multiple functions with the same name but different parameters.

Advantages :-

- Improves readability
- Reduces code duplication
- Easier maintenance

Program :-


```
#include <iostream>
using namespace std;
```

```
float area (float r) {
    return 3.14 * r * r;
}
```

```
int area (int s) {
    return s * s;
}
```

```
float area (float b, float h) {
    return 0.5 * b * h;
}
```

```
int main () {
    cout << "Circle : " << area (2.5) << endl;
    cout << "Square : " << area (4)
    cout << "Triangle : " << area (5.0, 6.0) << endl;
    return 0;
}
```

Q.2 what is operator overloading? Write a program to add two complex numbers using operator overloading.

Ans operator overloading allows operators to work with user-defined types.

Program :-


```
#include <iostream>
using namespace std;

class complex {
    int real, imag;
public:
    complex (int r=0, int i=0) {
        real = r;
        imag = i;
    }

    complex operator + (complex c) {
        return complex (real + c.real, imag + c.imag);
    }

    void display () {
        cout << real << "+" << imag << "i" << endl;
    }
};

int main () {
    complex c1 (2,3), c2 (4,5);
    complex c3 = c1 + c2;
    c3.display ();
    return 0;
}
```

Q.3 Design a class student in C++ with the following data members: roll number, name, and marks in three subjects.

Implement:

A default constructor to take input from the user and display

student details, average marks and percentage.

Ans # include <iostream>
using namespace std;

```
class Student {  
    int roll;  
    string name;  
    float m1, m2, m3;
```

public:

Student () {

cout << "Enter roll: ";

cin >> roll;

cout << "Enter name: ";

cin >> Name;

cout << "Enter marks (3 subjects): ";

cin >> m1 >> m2 >> m3;

display ();

}

void display () {

float avg = (m1 + m2 + m3) / 3;

float percentage = avg;

cout << "In Roll: " << roll << endl;

cout << "Name: " << name << endl;

cout << "Average: " << avg << endl;

cout << "Percentage: " << percentage << "%" << endl;

};

int main () {

Student s1;

return 0;

}